

Q1 - 25 July - Shift 1

Let $A = \begin{bmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}$ and $B = A - I$. If $\omega = \frac{\sqrt{3}i - 1}{2}$,

then the number of elements in the set $\{n \in \{1, 2, \dots, 100\} : A^n + (\omega B)^n = A + B\}$ is equal to _____.

Space for your notes:

Q2 - 25 July - Shift 2

Let $A = \begin{bmatrix} 1 & a & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix}$, $a, b \in \mathbb{R}$. If for some $n \in \mathbb{N}$,

$A^n = \begin{bmatrix} 1 & 48 & 2160 \\ 0 & 1 & 96 \\ 0 & 0 & 1 \end{bmatrix}$ then $n + a + b$ is equal to _____.

Space for your notes:

Q3 - 26 July - Shift 1

Let A be a 2×2 matrix with $\det(A) = -1$ and $\det((A + I)(\text{Adj}(A) + I)) = 4$. Then the sum of the diagonal elements of A can be :

(A) -1

(B) 2

(C) 1

(D) $-\sqrt{2}$

Space for your notes:

Q4 - 26 July - Shift 2

Let $A = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $B = \begin{bmatrix} 9^2 & -10^2 & 11^2 \\ 12^2 & 13^2 & -14^2 \\ -15^2 & 16^2 & 17^2 \end{bmatrix}$, then the

value of $A^T B A$ is:

- (A) 1224 (B) 1042 (C) 540 (D) 539

Space for your notes:

Q5 - 26 July - Shift 2

The number of matrices $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where $a, b, c, d \in \{-1, 0, 1, 2, 3, \dots, 10\}$, such that $A = A^{-1}$, is _____.

Space for your notes:

Q6 - 27 July - Shift 1

Let $A = \begin{pmatrix} 1 & 2 \\ -2 & -5 \end{pmatrix}$. Let $\alpha, \beta \in \mathbb{R}$ be such that

$\alpha A^2 + \beta A = 2I$. Then $\alpha + \beta$ is equal to -

- (A) -10 (B) -6
(C) 6 (D) 10

Space for your notes:

Q7 - 27 July - Shift 1

Let S be the set containing all 3×3 matrices with entries from $\{-1, 0, 1\}$. The total number of matrices $A \in S$ such that the sum of all the diagonal elements of $A^T A$ is 6 is _____.

Space for your notes:

Q8 - 27 July - Shift 2

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Let $A = \begin{pmatrix} 4 & -2 \\ \alpha & \beta \end{pmatrix}$

If $A^2 + \gamma A + 18I = O$, then $\det(A)$ is equal to

_____.

- (A) -18 (B) 18 (C) -50 (D) 50

Space for your notes:

Q9 - 27 July - Shift 2

Consider a matrix $A = \begin{bmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ \beta + \gamma & \gamma + \alpha & \alpha + \beta \end{bmatrix}$,

where α, β, γ are three distinct natural numbers.

If $\frac{\det(\text{adj}(\text{adj}(\text{adj}(\text{adj}A))))}{(\alpha - \beta)^{16}(\beta - \gamma)^{16}(\gamma - \alpha)^{16}} = 2^{32} \times 3^{16}$, then the

number of such 3 - tuples (α, β, γ) is _____.

Space for your notes:

Q10 - 28 July - Shift 1

Let the matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ and the matrix

$B_0 = A^{49} + 2A^{98}$. If $B_n = \text{Adj}(B_{n-1})$ for all $n \geq 1$, then $\det(B_4)$ is equal to :

- (A) 3^{28} (B) 3^{30} (C) 3^{32} (D) 3^{36}

Space for your notes:

Q11 - 28 July - Shift 1

Let $A = \begin{bmatrix} 1 & -1 \\ 2 & \alpha \end{bmatrix}$ and $B = \begin{bmatrix} \beta & 1 \\ 1 & 0 \end{bmatrix}$, $\alpha, \beta \in \mathbb{R}$. Let α_1 be the value of α which satisfies $(A+B)^2 = A^2 + \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ and α_2 be the value of α which satisfies $(A+B)^2 = B^2$. Then $|\alpha_1 - \alpha_2|$ is equal to _____.

Space for your notes:

Q12 - 28 July - Shift 2

Let A and B be any two 3×3 symmetric and skew symmetric matrices respectively. Then which of the following is **NOT** true?

Space for your notes:

- (A) $A^4 - B^4$ is a symmetric matrix
- (B) $AB - BA$ is a symmetric matrix
- (C) $B^5 - A^5$ is a skew-symmetric matrix
- (D) $AB + BA$ is a skew-symmetric matrix

Q13 - 29 July - Shift 1

Let A and B be two 3×3 non-zero real matrices such that AB is a zero matrix. Then

Space for your notes:

- (A) The system of linear equations $AX = 0$ has a unique solution
- (B) The system of linear equations $AX = 0$ has infinitely many solutions
- (C) B is an invertible matrix
- (D) $\text{adj}(A)$ is an invertible matrix

Q14 - 29 July - Shift 1

The number of matrices of order 3×3 , whose entries are either 0 or 1 and the sum of all the entries is a prime number, is _____.

Space for your notes:

Q15 - 29 July - Shift 2

Which of the following matrices can NOT be obtained from the matrix $\begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$ by a single elementary row operation?

Space for your notes:

(A) $\begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} -1 & 2 \\ -2 & 7 \end{bmatrix}$ (D) $\begin{bmatrix} -1 & 2 \\ -1 & 3 \end{bmatrix}$

Q16 - 29 July - Shift 2

Let $x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix}$. For $k \in \mathbb{N}$, if

Space for your notes:

$X' A^k X = 33$, then k is equal to:

Answer Key

Q1 (17)

Q2 (24)

Q3 (B)

Q4 (D)

Q5 (50)

Q6 (D)

Q7 (5376)

Q8 (B)

Q9 (42)

Q10 (C)

Q11 (2)

Q12 (C)

Q13 (B)

Q14 (282)

Q15 (C)

Q16 (10)

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Q1 (17)

$$A = \begin{bmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix} \Rightarrow A^2 = A \Rightarrow A^n = A.$$

$$\forall n \in \{1, 2, \dots, 100\}$$

$$\text{Now, } B = A - I = \begin{bmatrix} 1 & -1 & -1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{bmatrix}$$

$$B^2 = -B$$

$$\Rightarrow B^3 = -B^2 = B$$

$$\Rightarrow B^5 = B$$

$$\Rightarrow B^{99} = B$$

$$\text{Also, } \omega^{3k} = 1$$

So, n = common of $\{1, 3, 5, \dots, 99\}$ and $\{3, 6, 9, \dots, 99\} = 17$

Q2 (24)

(24)

Q3 (B)

$$\text{Let } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}; ad - bc = -1$$

$$|A + I| |\text{adj } A + I| = 4$$

$$\Rightarrow ad - bc + a + d + 1 = 2 \text{ or } -2$$

$$a + d = 2 \text{ or } -2$$

Q4 (D)

$$A'BA = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 9^2 & -10^2 & 11^2 \\ 12^2 & 13^2 & -14^2 \\ -15^2 & 16^2 & 17^2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 9^2 + 12^2 - 15^2 & -10^2 + 13^2 + 16^2 & 11^2 - 14^2 + 17^2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 9^2 + 12^2 - 15^2 & -10^2 + 13^2 + 16^2 & 11^2 - 14^2 + 17^2 \end{bmatrix}$$

$$= \begin{bmatrix} 539 \end{bmatrix}$$

Q5 (50)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\text{Given } A = A^{-1}$$

$$\therefore A^2 = A \cdot A^{-1} = I$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

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$$\begin{bmatrix} ac + cd & bc + d^2 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix}$$

Hints and Solutions

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$$\therefore a^2 + bc = 1 \quad \dots(1)$$

$$ab + bd = 0 \quad \dots(2)$$

$$ac + cd = 0 \quad \dots(3)$$

$$bc + d^2 = 1 \quad \dots(4)$$

(1) - (4) gives

$$a^2 - d^2 = 0$$

$$\Rightarrow (a + d) = 0 \text{ or } a - d = 0$$

Case - I

$$a + d = 0 \Rightarrow (a, d) = (-1, 1), (0, 0), (1, -1)$$

$$(a) (a, d) = (-1, 1)$$

$\therefore$ from equation (1)

$$1 + bc = 1 \Rightarrow bc = 0$$

$b = 0$ C = 12 possibilities

$c = 0$ b = 12 possibilities

but (0, 0) is repeated

$$\therefore 2 \times 12 = 24$$

$$24 - 1 (\text{repeated}) = 23 \text{ pairs}$$

$$(b) (a, d) = (1, -1) \Rightarrow bc = 0 \rightarrow 23 \text{ pairs}$$

$$(c) (a, d) = (0, 0) \Rightarrow bc = 1$$

$$\Rightarrow (b, c) = (1, 1) \text{ \& } (-1, -1), 2 \text{ pairs}$$

Case - II

$$a = d$$

from (2) and (3)

$$a \neq 0 \text{ then } b = c = 0$$

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$$a^2 = 1$$

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$(a, d) = (1, 1), (-1, -1) \rightarrow 2$ pairs

$$\therefore \text{Total} = 23 + 23 + 2 + 2 \\ = 50 \text{ pairs}$$

Q6 (D)

Characteristic equation of matrix A

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 2 \\ -2 & -5-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^2 + 4\lambda = 1$$

$$\Rightarrow A^2 + 4A = I$$

$$\Rightarrow 2A^2 + 8A = 2I \quad \dots\dots\dots (1)$$

Given that $\alpha A^2 + \beta A = 2I \quad \dots\dots\dots (2)$

Comparing equation (1) & (2) we get

$$\alpha = 2, \quad \beta = 8$$

$$\therefore \alpha + \beta = 10$$

Ans. (D) (10)

Q7 (5376)

$$\text{Tr}(AA^T) = 6$$

$$AA^T = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Now given $a^2 + d^2 + g^2 + b^2 + e^2 + h^2 + c^2 + f^2 + i^2 = 6$

$$= {}^9C_3 \times 2^6$$

$$= 5376$$

Q8 (B)

The characteristic equation for A is $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 4 - \lambda & -2 \\ \alpha & \beta - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (4 - \lambda)(\beta - \lambda) + 2\alpha = 0$$

$$\Rightarrow \lambda^2 - (\beta + 4)\lambda + 4\beta + 2\alpha = 0$$

Put $\lambda = A$

$$A^2 - (\beta + 4)A + (4\beta + 2\alpha)I = 0$$

On comparison

$$-9(\beta + 4) = \gamma \quad \& \quad 4\beta + 2\alpha = 18$$

$$\text{and } |A| = 4\beta + 2\alpha = 18$$

Q9 (42)

$$A = \begin{bmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ \beta + \gamma & \gamma + \alpha & \alpha + \beta \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_1$$

$$\Rightarrow |A| = |\alpha + \beta + \gamma| \begin{vmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ 1 & 1 & 1 \end{vmatrix}$$

$$\Rightarrow |A| = (\alpha + \beta + \gamma)(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$$

$$\therefore |\text{adj } A| = |A|^{n-1}$$

$$|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$$

$$|\text{adj}(\text{adj}(\text{adj}(\text{adj } A)))| = |A|^{(n-1)^4} = |A|^{2^4} = |A|^{16}$$

$$\therefore (\alpha + \beta + \gamma)^{16} = 2^{32} \cdot 3^{16}$$

$$\Rightarrow (\alpha + \beta + \gamma)^{16} = (2^2 \cdot 3)^{16} = (12)^{16}$$

$$\Rightarrow \alpha + \beta + \gamma = 12$$

$$\therefore \alpha, \beta, \gamma \in \mathbb{N}$$

$$(\alpha - 1) + (\beta - 1) + (\gamma - 1) = 9$$

$$\text{number all tuples } (\alpha, \beta, \gamma) = {}^{11}C_2 = 55$$

$$1 \text{ case for } \alpha = \beta = \gamma$$

$$\& 12 \text{ case when any two of these are equal}$$

$$\text{So, No. of distinct tuples } (\alpha, \beta, \gamma)$$

$$= 55 - 13 = 42$$

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Q10 (C)

$$A^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$a \leftrightarrow R_2$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$B_0 = A^{49} + 2A^{98}$$

$$= A + 2I$$

$$B_n = \text{Adj}(B_{n-1})$$

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$$B_4 = \text{Adj}(\text{Adj}(\text{Adj}(\text{Adj } B_0)))$$

$$= |B_0|^{(n-1)^4}$$

$$= |B_0|^{16}$$

$$B_0 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

$$= 2(4-0) - 1(0-1)$$

$$= 9$$

$$B_4(9)^{16} = (3)^{32}$$

Q11 (2)

$$A + B = \begin{bmatrix} \beta + 1 & 0 \\ 3 & \alpha \end{bmatrix}$$

$$(A + B)^2 = \begin{bmatrix} \beta + 1 & 0 \\ 3 & \alpha \end{bmatrix} \begin{bmatrix} \beta + 1 & 0 \\ 3 & \alpha \end{bmatrix}$$

$$= \begin{bmatrix} (\beta + 1)^2 & 0 \\ 3(\beta + 1) + 3\alpha & \alpha^2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & -1 \\ 2 & \alpha \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & \alpha \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -1 - \alpha \\ 2 + 2\alpha & \alpha^2 - 2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 1 & -\alpha + 1 \\ 2\alpha + 4 & \alpha^2 \end{bmatrix} = \begin{bmatrix} (\beta + 1)^2 & 0 \\ 3(\alpha + \beta + 1) & \alpha^2 \end{bmatrix}$$

$$\boxed{\alpha = 1} = \alpha_1$$

$$B^2 = \begin{bmatrix} \beta & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \beta & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \beta^2 + 1 & \beta \\ \beta & 1 \end{bmatrix} = \begin{bmatrix} (\beta + 1)^2 & 0 \\ 3(\beta + 1) + 3\alpha & \alpha^2 \end{bmatrix}$$

$$\therefore \beta = 0, \alpha = -1 = \alpha_2$$

$$|\alpha_1 - \alpha_2| = |1 - (-1)| = 2$$

Q12 (C)

Given that $A^T = A$, $B^T = -B$

(A) $C = A^4 - B^4$

$$C^T = (A^4 - B^4)^T = (A^4)^T - (B^4)^T = A^4 - B^4 = C$$

(B) $C = AB - BA$

$$\begin{aligned} C^T &= (AB - BA)^T = (AB)^T - (BA)^T \\ &= B^T A^T - A^T B^T = -BA + AB = C \end{aligned}$$

(C) $C = B^5 - A^5$

$$C^T = (B^5 - A^5)^T = (B^5)^T - (A^5)^T = -B^5 - A^5$$

(D) $C = AB + BA$

$$\begin{aligned} C^T &= (AB + BA)^T = (AB)^T + (BA)^T \\ &= -BA - AB = -C \end{aligned}$$

$\therefore$ Option C is not true.

Q13 (B)

$$AB = 0 \Rightarrow |AB| = 0$$

$$\begin{array}{c} |A| \quad |B| = 0 \\ \swarrow \quad \searrow \\ |A| = 0 \quad |B| = 0 \end{array}$$

$$|A| = 0 \quad |B| = 0$$

If $|A| \neq 0$, $B = 0$ (not possible)

If $|B| \neq 0$, $A = 0$ (not possible)

Hence $|A| = |B| = 0$

$\Rightarrow AX = 0$ has infinitely many solutions

Q14 (282)

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} a_{ij} \in \{0,1\}$$

$$\sum a_{ij} = 2, 3, 5, 7$$

$$\text{Total matrix} = {}^9C_2 + {}^9C_3 + {}^9C_5 + {}^9C_7$$

$$= 282$$

Q15 (C)

$$A = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$$

$$(1) R_1 \rightarrow R_1 + R_2; \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} \text{ possible}$$

$$(2) R_1 \leftrightarrow R_2; \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \text{ possible}$$

(3) Option is not possible

$$(4) R_2 \rightarrow R_2 + 2R_1; \begin{bmatrix} -1 & 2 \\ -1 & 3 \end{bmatrix} \text{ possible}$$

Q16 (10)

$$X = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}; A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix}$$

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Hints and Solutions

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$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix}^k \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 33$$

$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 33$$

$$\text{As } A^2 = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 12 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^8 = \begin{bmatrix} 1 & 0 & 24 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{10} = \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 24 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 30 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{for } K \rightarrow \text{Even } A^K = \begin{bmatrix} 1 & 0 & 3K \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$X^T A^K X = 33 \text{ (This is not correct)}$$

$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3K \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 3K+1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 3K+3$$

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Hints and Solutions

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$$\therefore 3K + 3 = 33 \therefore K = 10$$

But it should be dropped as 33 is not matrix

If K is odd

$$X^T A^K X = 33$$

$$X^T A A^{K-1} X = 33$$

$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3k-3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 33$$

$$\begin{bmatrix} -1 & 3 & 8 \end{bmatrix} \begin{bmatrix} 3k-2 \\ 1 \\ 1 \end{bmatrix} = [33]$$

$$[-3k + 13] = [33]$$

$$k = 20/3 \text{ (not possible)}$$